

KALAMATA, Greece

WMO: 167260

Lat: 37.068N

Lon: 22.026E

Elev: 8

StdP: 101.23

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.9	(c) 2.2	(d) -5.9	(e) 2.3	(f) 5.7	(g) -3.7	(h) 2.8	(i) 6.0	(j) 9.8	(k) 14.0	(l) 8.5	(m) 13.4	(n) 2.1	(o) 310	(p) 0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.6	(c) 34.9	(d) 21.1	(e) 33.1	(f) 21.4	(g) 32.0	(h) 21.3	(i) 24.1	(j) 29.7	(k) 23.5	(l) 29.4	(m) 22.8	(n) 29.1	(o) 4.1	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	16.9	27.0	21.7	16.4	26.8	20.9	15.6	26.4	72.7	29.5	69.9	29.6	67.6	29.1	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.9	6.0	-2.0	39.5	1.4	2.1	-3.0	40.9	-3.9	42.1	-4.6	43.3	-5.7	44.8
			-3.1	25.6	1.5	1.1	-4.2	26.4	-5.1	27.0	-6.0	27.6	-7.1	28.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	10.1	10.5	12.1	14.9	19.0	23.2	25.7	25.9	22.8	19.1	14.9	11.4
	DBStd	6.13	2.52	2.54	2.36	2.31	2.32	2.38	1.91	1.84	2.15	2.33	2.58	2.66
	HDD10.0	77	29	22	7	1	0	0	0	0	0	0	2	15
	HDD18.3	1133	255	220	192	106	20	1	0	0	1	19	104	214
	CDD10.0	2812	32	36	73	146	279	396	486	492	383	281	150	59
	CDD18.3	826	0	0	0	2	41	146	227	234	133	42	2	0
	CDH23.3	8743	0	0	2	31	386	1558	2647	2652	1174	281	13	0
	CDH26.7	3599	0	0	0	2	72	584	1258	1282	372	29	0	0
Wind	WSAvg	2.2	2.1	2.3	2.2	2.1	2.2	2.6	2.7	2.6	2.3	1.9	2.0	2.0
Precipitation	PrecAvg	715	103	83	60	40	24	15	8	14	45	68	124	131
	PrecMax	1018	233	195	133	93	63	83	24	40	137	153	308	364
	PrecMin	403	8	8	0	5	4	0	0	0	0	3	8	14
	PrecStd	132	61	46	28	24	15	20	7	13	35	44	70	74
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	20.0	22.4	26.2	31.0	36.0	38.0	37.2	34.0	29.1	25.0	21.0
		MCWB	14.2	13.4	14.2	16.1	18.6	20.7	20.8	21.0	20.2	19.1	18.1	16.0
	2%	DB	17.8	18.1	20.2	24.0	28.2	33.0	34.8	34.9	31.2	27.1	23.1	19.2
		MCWB	13.4	12.9	14.0	15.4	18.3	20.6	21.3	21.2	20.6	19.2	17.3	14.7
	5%	DB	16.5	17.1	19.0	22.2	26.9	31.2	33.1	33.1	30.0	26.0	21.9	18.1
		MCWB	12.8	12.7	13.4	14.9	18.1	20.6	21.7	21.5	20.7	18.9	16.7	14.0
	10%	DB	15.2	15.9	17.9	21.0	25.2	30.0	31.9	32.0	28.9	25.0	20.6	16.9
		MCWB	11.7	12.1	13.0	14.7	17.7	20.6	21.5	21.6	20.5	18.6	15.9	13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	15.1	16.0	17.7	20.9	23.7	24.8	25.4	24.0	22.3	19.7	17.4
		MCDB	17.7	17.8	20.0	23.2	27.2	29.4	30.1	30.4	28.5	25.4	23.1	19.8
	2%	WB	14.1	14.0	15.0	16.6	19.7	22.7	24.0	24.4	22.9	21.0	18.4	15.6
		MCDB	16.5	17.1	18.9	21.8	25.7	29.0	29.9	29.9	27.9	25.0	21.8	18.3
	5%	WB	13.2	13.1	14.2	15.8	19.1	22.0	23.4	23.6	22.1	20.1	17.4	14.6
		MCDB	15.8	16.2	17.8	20.7	24.9	28.8	29.7	29.5	27.5	24.4	20.7	17.4
	10%	WB	12.2	12.3	13.5	15.2	18.4	21.2	22.7	22.9	21.4	19.3	16.4	13.5
		MCDB	14.9	15.4	17.0	19.7	24.0	28.2	29.5	29.2	26.8	23.8	19.6	16.1
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.6	10.6	11.3	12.1	12.5	12.7	12.6	11.6	11.1	10.3	9.6
		MCDBR	10.8	11.0	12.3	13.3	14.2	14.5	14.4	14.2	13.0	12.1	11.5	10.8
		MCWBR	7.0	6.9	7.2	6.9	6.4	6.2	6.1	6.1	5.9	6.2	6.8	6.9
	5% WB	MCDBR	9.2	10.0	11.2	11.8	12.5	13.1	12.8	12.5	11.7	10.9	10.5	9.1
		MCWBR	6.6	6.8	7.2	6.8	6.4	6.4	6.4	6.4	6.2	6.3	6.8	6.6
Clear-Sky Solar Irradiance	taub	0.359	0.390	0.437	0.475	0.466	0.439	0.444	0.457	0.440	0.417	0.388	0.367	
	taud	2.398	2.283	2.138	2.054	2.139	2.268	2.259	2.221	2.262	2.291	2.342	2.380	
	Ebn at Noon	817	832	822	812	825	846	838	819	811	791	773	778	
	Edh at Noon	90	113	143	165	154	135	135	138	125	111	95	87	
All-Sky Solar Radiation	RadAvg	2.18	2.98	4.38	5.68	6.69	7.56	7.75	6.92	5.33	3.73	2.56	1.96	
	RadStd	0.23	0.27	0.32	0.45	0.31	0.37	0.22	0.20	0.28	0.27	0.23	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.29	N/A	N/A	N/A	-0.78	-0.97	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+0.29	N/A	N/A	+0.60	N/A	N/A	N/A	-72	+85	N/A

Nomenclature: See separate page